The Hidden Tax on Working Families

Findings

The National Database of Childcare Prices (NDCP) covers the years 2008 through 2018 and encompasses more than 3,000 counties across all 50 states. The dataset captures weekly median and 75th-percentile childcare costs for center-based and family home-based care across four standardized age groups: infants, toddlers, preschool-age, and school-age children.

Analysis of the data revealed four central findings. First, infant center-based care is the most expensive care category in almost every county, at a national median of $153.60 per week in 2018. This is $28.60 more per week than preschool-age care, which translates to over $1,487 more annually. Second, when annual infant care costs are expressed as a percentage of median household income using inflation-adjusted 2018 dollars, the national median burden rose from approximately 13.5% in 2008 to 16.1% in 2018. This is more than double the 7% threshold the U.S. Department of Health and Human Services defines as affordable. Third, at least 75% of U.S. counties exceed that affordability threshold, and the five most expensive states show persistent upward pressure with no state returning to its 2008 burden level. Fourth, a scatter plot of county-level infant care cost against female labor force participation rates for mothers of children under six reveals a downward trend: counties with higher childcare costs show measurably lower women's workforce participation, confirming that the childcare affordability crisis carries direct labor market consequences.

Assumptions

Several assumptions were made during the analysis. The NDCP uses predictive mean matching imputation to fill gaps where county-level survey data was unavailable, and the analysis assumes that these imputed values are sufficiently reliable for trend and comparative purposes. The dataset covers only licensed and regulated childcare providers, which likely understates the affordability burden in lower-income communities. All time-series income comparisons use the MHI_2018 inflation-adjusted variable rather than nominal household income. Finally, the standardized age group definitions were assumed to be consistent enough across states to support cross-county comparison.

Items That Still Need Clarification

Two open questions from Milestone 1 were partially resolved during the analysis. Imputation concentration by state was examined: states including FL, MN, NV, NC, PA,

and SD showed higher imputation rates for key price variables. The single-mother income comparator question was also explored: infant care expressed as a percentage of female median earnings (FME_2018) produces burdens significantly higher than the household income ratio confirming that the standard affordability metric systematically understates the burden on single-mother households. One remaining clarification is whether school-age pricing data is reliable enough across all counties to include in future analyses, given its comparatively sparse coverage.

Direction of Story

The project tells a single, evidence-backed story: childcare in America has become a financial barrier for working families, and this burden falls disproportionately on women. The narrative arc moves from a human-scale framing (what a new parent pays each week) through national trend evidence (costs rising faster than incomes), geographic and state-level comparison, and the workforce participation consequence, closing with a direct call to action: employers and state legislators should treat childcare subsidies as workforce infrastructure rather than discretionary social programs. This framing was maintained consistently across all three mediums, with each medium carrying a different portion of the argument suited to its format and audience.

Target Audience

The primary audience is human resources and benefits decision-makers at mid-to-large employers, as well as state-level policy advisors. These individuals have direct influence over benefits structures and funding decisions that most immediately affect working parents. They are likely sympathetic to a workforce framing of the childcare problem, rather than a social-welfare framing, which informed every design and messaging decision across the three mediums.

Medium: PowerPoint Presentation

The PowerPoint deck was selected as the primary narrative medium because it supports a sequential, presenter-guided argument. The deck uses five chart slides that require verbal or speaker-note context to interpret correctly. Charts requiring the most explanation were reserved exclusively for the PowerPoint and not duplicated in other mediums.

Medium: Infographic

The infographic was selected to serve working parents and advocacy audiences who need the core message communicated in under thirty seconds without data

literacy. Two charts were created: the weekly cost distribution histogram and the female labor force participation quartile bar chart. The infographic is formatted for print and social media sharing, making it the most widely distributable of the three mediums and the one most likely to reach audiences beyond the primary HR and policy target.

Medium: Tableau Dashboard

The Tableau dashboard was selected for policy researchers and legislative staff who want to explore the data for their own geography rather than follow a fixed narrative. Interactivity is the dashboard's defining value: a shared year filter connects a county-level choropleth map, a state-filtered affordability trend line, and a state ranking bar chart, allowing the audience to move fluidly between geographic, temporal, and comparative perspectives.

Design Decisions

Color was used functionally and intentionally. The PowerPoint uses a dark navy background on title and closing slides with white content slides in between, creating a visual 'sandwich' structure that distinguishes section transitions. The infographic uses a strict top-to-bottom zone structure with section label bars to make the layout scannable without a presenter. The Tableau dashboard uses a dark navy background to visually distinguish it from the white-background PowerPoint and infographic, reinforcing that it serves a different exploratory purpose.

Data Transformations

The primary transformations applied to the raw dataset were the creation of derived variables: annual childcare cost (weekly price × 52), affordability ratio (annual cost ÷ MHI_2018 × 100), infant cost quartile classifications, and infant-to-preschool cost gaps. All income comparisons use the inflation-adjusted MHI_2018 variable rather than nominal household income to ensure that trend findings reflect genuine real cost increases rather than nominal dollar growth. No county records were removed from the underlying dataset without disclosure; where imputed values were used, they are noted and the imputation flag columns were documented. Lastly, three pre-aggregated summary files were also created to support accurate Tableau visualizations: a state-level summary for the ranking chart, a year-level summary for the trend line, and a single-row KPI summary for the dashboard tiles.

Legal and Regulatory Guidelines

The NDCP is a federal open dataset published by the U.S. Department of Labor Women's Bureau under standard government data publication policies. It is publicly available for research and analysis without restriction. No personally identifiable information is present in the dataset. All data is aggregated to the county-year level. There are no legal or licensing barriers to using, analyzing, or visualizing this data in academic or professional contexts.

Data Sourcing and Credibility

The NDCP was produced by ICF International under contract with the U.S. Department of Labor Women's Bureau. The dataset underwent peer review and is accompanied by a published technical report detailing the methodology, imputation procedures, and variable definitions. Its federal origin and OMB-approved collection process make it one of the most credible publicly available sources for county-level childcare pricing data in the United States.

Mitigation of Ethical Implications

Four mitigation steps were applied. Imputed values are disclosed where they represent a material share of a key variable. The regulated-provider limitation is stated explicitly in the data overview slide and infographic source note, so the audience understands the dataset does not capture informal care arrangements. Correlational findings are framed explicitly as associations rather than causal relationships in all three mediums. The quartile bar chart shows a slight positive relationship between childcare cost and female workforce participation, which is counterintuitive. This is explained by the urban income premium (high-cost counties are predominantly urban with higher wages) and is accompanied by an explicit explanation in the presentation to prevent the audience from drawing an incorrect conclusion.

What Would Be Done Differently

The most significant change for future projects would be to finalize the visual differentiation plan before building any individual medium. In this project, the duplicate chart between the PowerPoint and infographic was discovered during Milestone 3 review rather than caught during planning. Mapping each visualization to exactly one medium at the outset would prevent this. Secondly, the infographic would be started earlier in the process, as it proved more technically complex than anticipated and would have benefited from more iteration time.

What Was Most Enjoyable

This has been an insightful term project compared to previous classes. I especially enjoyed working with Tableau for the first time and getting a better understanding on how to use this tool. The textbooks have provided a good foundation and guidance on how to share our data story.

Reference

ICF. (2024, September). *National database on childcare prices: Technical report*. U.S. Department of Labor, Women's Bureau. https://www.dol.gov/sites/dolgov/files/WB/National-Database-of-Childcare-Prices-Technical-Report.pdf